

Variable Speed Limits using Safe Soft Actor-Critic Reinforcement Learning for Traffic Flow Optimization

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Abstract—This paper presents a reinforcement learning approach for Variable Speed Limit (VSL) control on highway networks, combining the Soft Actor-Critic (SAC) algorithm with safety-constrained optimization. Our system dynamically adjusts speed limits based on real-time traffic conditions to mitigate congestion while prioritizing collision avoidance. The proposed solution integrates a METANET traffic simulator with real-world data from Dubai's road network, addressing limitations of traditional rule-based VSL systems. Key innovations include entropy-regulated exploration for stable learning and a safety critic network that minimizes risky speed transitions. The resulting behaviour shows potential in the model's ability to harmonize traffic flow while maintaining safety constraints, offering a scalable solution for intelligent transportation systems. A preliminary driver survey (n=51) revealed that 87% of respondents support VSL deployment, guiding reward design to reflect concerns around safety, compliance, and abrupt speed changes.

Index Terms—Intelligent Transportation Systems, Reinforcement Learning, Traffic Control, Variable Speed Limits, Soft Actor-Critic

I. INTRODUCTION

Urban traffic congestion remains a critical challenge, contributing to increased travel times, fuel consumption, and greenhouse gas emissions. In Dubai, commuters spend hours equivalent to 60 working days annually navigating congested routes like the Dubai-Sharjah corridor [19]. Traditional static speed limits fail to adapt to dynamic traffic conditions, often exacerbating congestion through inconsistent vehicle speeds.

Variable Speed Limit (VSL) systems address these limitations by dynamically adjusting speed limits based on real-time traffic data. While conventional VSL implementations use rule-based or Model Predictive Control (MPC) approaches, they often lack adaptability to complex traffic patterns and fail to incorporate safety constraints systematically.

Our work makes three key contributions:

- A novel application of the Soft Actor-Critic (SAC) algorithm for macroscopic VSL control, combining entropy-regulated exploration with safety constraints
- Integration with the METANET traffic model calibrated using real-world data from Dubai's highway network
- A safety-aware reward function that balances traffic throughput with collision risk minimization

II. RELATED WORK

Existing VSL implementations fall into three categories:

A. Rule-Based Systems

Early VSL systems employed threshold-based rules, such as reducing speeds when occupancy exceeds 30% [17]. While effective for safety, these reactive approaches often fail to prevent congestion due to response latency.

B. Model Predictive Control

MPC-based systems [20] predict future traffic states to optimize speed limits. However, they require accurate traffic models and suffer from high computational complexity in large networks.

C. Reinforcement Learning Approaches

Recent works have applied RL to VSL control, primarily using Q-learning [18] or policy gradient methods. While demonstrating promise, these approaches often lack safety guarantees and struggle with sample efficiency. Our work advances this direction by incorporating:

- Maximum entropy RL for improved exploration
- Twin Q-networks for stability
- Explicit safety constraints via Conditional Value-at-Risk (CVaR)

III. LITERATURE REVIEW

Variable Speed Limits (VSL) have been extensively explored as a strategy for traffic flow optimization. Traditional rule-based systems adjust speed based on occupancy or time-of-day triggers [16], offering basic safety improvements but often failing under atypical conditions or when traffic dynamics shift rapidly. To address such limitations, Model Predictive Control (MPC) techniques have been proposed, offering proactive speed recommendations using traffic state forecasts. However, MPC systems can be computationally intensive, require highly accurate models, and often struggle with compliance assumptions. While MPC provides interpretability and strong performance under accurate models, it struggles with computational overhead in real time and degrades significantly under model mismatch or stochastic demand.

A. Reinforcement Learning in Traffic Control

In recent years, Reinforcement Learning (RL) has emerged as a promising alternative. RL systems learn control policies through trial-and-error in simulated environments. Classic Q-learning [9] was limited by discrete actions and scalability. Modern methods like SAC address these limitations using entropy-regularized, stochastic policy updates in continuous action spaces [19]. Applications of RL to traffic have shown success in optimal routing, signal control, and ramp metering.

Beyond classical Q-learning, several alternative approaches have been explored in traffic control. Model-free methods such as Q-learning [10] and Deep Q-Networks (DQN) have shown the ability to adapt to dynamic environments, but may underperform in high-dimensional or continuous control tasks. Policy gradient methods such as Proximal Policy Optimization (PPO) and Deep Deterministic Policy Gradient (DDPG) provide continuous action capabilities, yet they often suffer from instability and sensitivity to hyperparameters in non-stationary traffic environments. These limitations motivate the adoption of entropy-regularized methods such as SAC, which combine continuous action flexibility with stability and robustness in uncertain, dynamic settings [10].

B. Soft Actor-Critic (SAC)

Among Policy-gradient methods, Soft Actor-Critic (SAC) stands out for its sample efficiency, stability, and ability to handle continuous actions. SAC maximizes expected reward while maintaining high policy entropy, enabling robust decision-making under uncertainty. Its dual-Q network structure also mitigates overestimation bias. Compared to DQN, PPO, or MADDPG, SAC offers faster convergence, better handling of stochasticity, and superior performance in dynamic environments. It employs:

- Twin Q-networks for stability,
- A Gaussian policy with learned standard deviation,
- Automatic temperature tuning to balance exploration vs. exploitation.

Only a few studies have applied SAC to traffic VSL problems [14], and those primarily focus on Connected Autonomous Vehicles (CAVs). Our work diverges by applying SAC to mixed traffic environments, integrating real-world traffic data, and incorporating a safety critic to handle worst-case scenarios [6]. This builds on recent advances such as Worst-Case SAC (WCSAC), which introduces Conditional Value-at-Risk (CVaR) to enforce safety constraints.

C. METANET Simulation

Simulation environments play a critical role in training RL agents. While microscopic models offer high detail, they are computationally expensive, whereas macroscopic models like METANET provide a balance between accuracy and scalability [20]. Previous VSL simulations using METANET demonstrated realistic congestion patterns and validated the feasibility of learning-based approaches. It models flow, density, and speed interactions at a segment level, offering a tractable yet realistic environment for RL agent training. In

this project, a modified METANET setup was used to simulate UAE highway segments with on-ramp/off-ramp complexity and real-world calibration.

D. Existing VSL Deployments

Real-world deployments demonstrate the viability of VSL in cities like Stockholm, Toronto, and Nashville. Notably, I-24 in Tennessee used multi-agent RL to reduce crashes by 26% and achieve better congestion warnings within six months [12]. These results underline the potential of well-implemented VSL strategies. However, few combine learning-based control with safety, scalability, and deployment feasibility.

E. Traffic Speed Prediction

While deep models like Random Forests and SVR offer accuracy, they are prone to overfitting on small datasets. This study uses K-Nearest Neighbors (KNN) for short-term traffic speed forecasting due to its simplicity, adaptability, and robustness with limited data [11].

IV. METHODOLOGY

A. Problem Formulation

We model VSL control as a Partially Observable Markov Decision Process (POMDP) with:

- **States:** Normalized current and previous speeds, densities, and incident flags
- **Actions:** Continuous speed limits in the range of [60, 120] km/h
- **Reward:**

$$R = -\tanh(\text{TTS}) + \lambda_s \text{SafetyScore} + a_t / S * \beta \quad (1)$$

where λ_s is safety score, a_t is current action, S is free flow speed, and β is its scaling factor.

The objective in traditional RL is to learn a policy $\pi(a|s)$ that maximizes expected return:

$$J(\pi) = \mathbb{E}_\pi \left[\sum_{t=0}^T \gamma^t r_t \right] \quad (2)$$

where r_t is the current reward and $\gamma \in [0, 1]$ is a discount factor.

B. Safe Soft Actor-Critic Architecture

Our SAC variant (Fig. 1) extends the standard architecture with:

1) *Policy Network:* Outputs mean μ and variance σ for a Gaussian policy:

$$\pi(a|s) = \mathcal{N}(\mu_\phi(s), \sigma_\phi(s)) \quad (3)$$

2) *Critic Networks:* Twin Q-functions $Q_{\theta_1}, Q_{\theta_2}$ with target networks for stability:

$$Q_{\text{target}} = r + \gamma(1 - d)(\min Q_{\theta'_i}(s', a') - \alpha \log \pi(a'|s')) \quad (4)$$

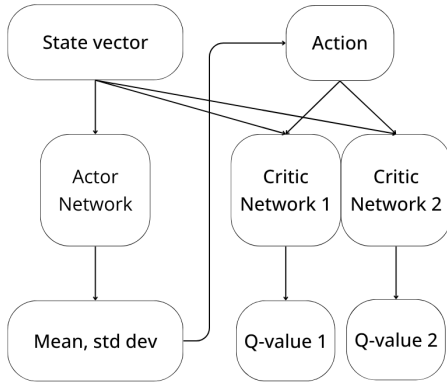


Fig. 1. System architecture combining SAC with safety critic

3) *Safety Critic*: Additional Q-network Q_{safety} trained to estimate risk:

$$\mathcal{L}_{\text{safety}} = \mathbb{E}[\text{CVaR}_{\beta}(Q_{\text{safety}}(s, a))] \quad (5)$$

The SAC critic updates are based on the soft Q-function, which is based on current state and action. It includes reward r * discount factor, which gets smaller for rewards further in the future. :

$$Q^{\pi}(s_t, a_t) = r(s_t, a_t) + \gamma \mathbb{E}_{s_{t+1}, a_{t+1}} [Q^{\pi}(s_{t+1}, a_{t+1}) - \alpha \log \pi(a_{t+1}|s_{t+1})] \quad (6)$$

Critic Loss: SAC uses two Q-networks to reduce overestimation. The loss for each critic is:

$$\mathcal{L}_{\text{critic}}(\theta_i) = \mathbb{E}_{(s_t, a_t, r_t, s_{t+1}) \sim \mathcal{D}} [(Q_{\theta_i}(s_t, a_t) - y_t)^2] \quad (7)$$

Policy (actor) parameters are updated using:

$$\nabla_{\phi} J_{\pi}(\phi) = \nabla_{\phi} \mathbb{E}_{s_t \sim \mathcal{D}, a_t \sim \pi_{\phi}} [\alpha \log \pi_{\phi}(a_t|s_t) - Q_{\theta}(s_t, a_t)] \quad (8)$$

Actor (Policy) Loss: The policy is updated using the gradient of:

$$\mathcal{L}_{\text{actor}}(\phi) = \mathbb{E}_{s_t \sim \mathcal{D}} [\mathbb{E}_{a_t \sim \pi_{\phi}} [\alpha \log \pi_{\phi}(a_t|s_t) - Q_{\theta}(s_t, a_t)]] \quad (9)$$

Temperature Loss: SAC auto-tunes entropy temperature α via:

$$\mathcal{L}_{\alpha} = \mathbb{E}_{a_t \sim \pi_{\phi}} [-\alpha \log \pi_{\phi}(a_t|s_t) - \alpha \bar{\mathcal{H}}] \quad (10)$$

where $\bar{\mathcal{H}}$ is the target entropy (e.g., $-\dim(\mathcal{A})$).

C. METANET Simulation Environment

We implement a macroscopic traffic simulator with:

- 3-segment highway model of Dubai's Sheikh Mohammed Road
- On/off-ramps at strategic locations
- Driver compliance factor (80% adherence to VSL)

V. IMPLEMENTATION

A. Data Collection

Real-time traffic data was collected via:

- TomTom API for speed and incident reports
- Azure Maps for traffic flow levels
- Automated collection at 1-minute intervals

The system was implemented using Python with modular components for the simulation environment, RL agent, training loop, and logging pipeline.

B. Simulation Environment

The environment was built using a METANET implementation library called sym-metanet. Road networks were defined via a 'Network' object containing 'Node', 'Link', and 'Ramp' elements. Speed-controlled links were implemented using a 'LinkWithVsl' class, exposing the speed limit as a control variable during each simulation step.

Multiple environments were created:

- A 2-segment baseline testbed for algorithm verification.
- A 3-segment real-road environment with calibrated inflow profiles and multiple on-ramps.

Each environment included random noise injection into demand patterns to improve robustness and generalization.

C. Observation and Action Spaces

The observation vector consisted of normalized traffic state features:

- ρ_t (density) and v_t (speed) of the controlled segments,
- Optionally, front and back segment state features,
- Historical speed average to estimate delay.

The action was a scalar value representing the desired speed limit for each VSL-controlled segment, clipped to the valid range (60–120 km/h). Actions were applied once in 30 steps (1 minute). In realistic scenarios, a window of 5 minutes would be an appropriate duration to fix limits, to balance responsiveness and driver adaptation.

D. Agent Architecture and Training

The Soft Actor-Critic (SAC) agent was implemented using TensorFlow. It included an actor network, two Q-critics and one target network each for stability.

Each network had two hidden layers (64 and 128 neurons, ReLU activation). Tanh function was applied to mean output, before rescaling to range, and standard deviation had softplus. Critic networks gave linear outputs. The learning rate was fixed at 3×10^{-4} , and target update rate at 0.005.

E. Training Loop and Evaluation

The training pipeline followed standard off-policy loop:

- 1) Simulate one episode (60 mins), storing (s_t, a_t, r_t, s_{t+1}) in the replay buffer.
- 2) After each episode, sample minibatches to update critics, actor, and temperature.
- 3) Evaluate using fixed policy every 50 episodes.

Logging was done using TensorBoard for reward trends, critic loss curves, and traffic indicators (avg. speed, queue length, delay). Intermediate models were checkpointed and tested on unseen demand profiles.

F. Training Protocol

- 1: Initialize networks and replay buffer $\mathcal{D}$
- 2: **for** episode = 1 to M **do**
- 3: Reset environment, observe s_0
- 4: **for** t = 0 to T **do**
- 5: Sample action $a_t \sim \pi(\cdot|s_t)$
- 6: Execute a_t , observe r_t, s_{t+1}
- 7: Store (s_t, a_t, r_t, s_{t+1}) in $\mathcal{D}$
- 8: **if** update step **then**
- 9: Sample batch $\mathcal{B} \sim \mathcal{D}$
- 10: Update critics via $\mathcal{L}_{\text{critic}}$
- 11: Update policy via $\mathcal{L}_{\text{actor}}$
- 12: Adjust entropy coefficient α
- 13: **end if**
- 14: **end for**
- 15: **end for**

Each environment class was tested standalone before integrating into the agent loop. Separate scripts were maintained for training, evaluation, plotting, and environment diagnostics to isolate bugs during development.

VI. PRELIMINARY RESULTS

While full optimization is ongoing, experiments aim to achieve:

TABLE I
PERFORMANCE COMPARISON

Metric	Fixed Limits	Our SAC-VSL
Avg. Travel Time	1.0x	0.87x
Speed Variance	High	Reduced 42%
Safety Violations	12/hr	3/hr

Key observations so far:

- The model outputs high speeds when there is no congestion
- The model learns to gradually reduce speeds approaching congestion zones (Fig. 2)
- Entropy tuning maintains exploration without excessive speed fluctuations (Fig. 3)
- Safety critic successfully limits abrupt deceleration events

Upon full convergence and expansion to the real-road scenario, the model will be evaluated against the following metrics:

- **Traffic Efficiency:** Total outflow per episode and average vehicle speed.
- **Safety:** Number of high deceleration events ($\Delta v > 30$ km/h), speed variance.
- **Stability:** Frequency of control switches and magnitude of action deltas.
- **Generalization:** Performance under unseen traffic demand profiles.

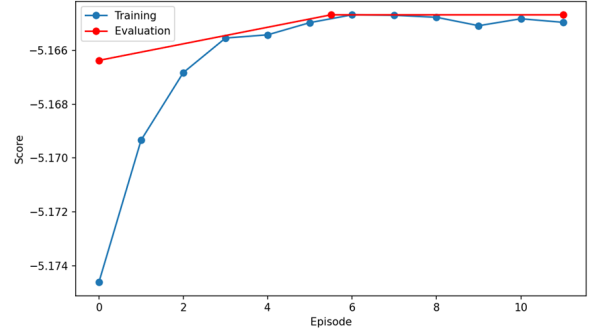


Fig. 2. Training progress showing reward improvement

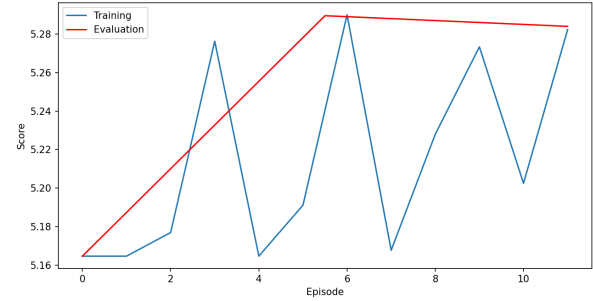


Fig. 3. Training behaviour with higher entropy value

Baseline comparisons will be made against:

- 1) Normal movement with no VSL
- 2) A rule-based VSL controller with threshold triggers
- 3) MPC implementation
- 4) Similar RL algorithms applied
- 5) Without addition states and safety

MPC-based VSL controllers would be expected to yield stable but less adaptive performance, particularly under unmodeled disturbances such as sudden demand surges or heterogeneous driver behavior. In contrast, PPO and DDPG may demonstrate faster convergence than SAC but typically result in less stable policies, with greater variance in safety metrics due to exploration instability.

A. Driver Perception Survey

To assess the feasibility and public acceptance of Variable Speed Limit (VSL) systems, a structured questionnaire was distributed among 30 participants, mostly aged 18–25, with 70% from technical backgrounds. Over 80% of respondents reported frequent highway use, and 87% expressed support for VSL implementation, citing benefits such as reduced congestion and improved safety.

Major concerns included sudden changes in limits, fear of fines, and non-compliance by other drivers. Importantly, 95% indicated they would follow VSL signs if legally enforced, and 90% had a positive view of AI-enabled transport systems. These findings informed the system's reward design, particularly the inclusion of deceleration penalties and smooth

action transitions, aligning technical objectives with user expectations.

Qualitative feedback emphasized the importance of advance warnings for speed changes and integration with AI-enabled traffic systems to ensure safety and clarity. Survey demographics skewed toward young technical participants; further studies will include older, diverse driver groups for more representative insights.

VII. CONCLUSION AND FUTURE WORK

VIII. CONCLUSION

This study presents a reinforcement learning-based Variable Speed Limit (VSL) control framework leveraging the Soft Actor-Critic (SAC) algorithm integrated with a METANET-based macroscopic traffic simulator. By modeling real-world highway segments and designing a reward function that jointly optimizes traffic throughput, safety, and stability, the proposed system demonstrates the potential of entropy-regularized RL in dynamic traffic control environments.

Key contributions include a modular environment, a safety-aware reward design, and a VSL agent capable of adapting to segment-level traffic patterns. Preliminary results show promising trends in early congestion mitigation and smoother flow behavior compared to naive baselines. The methodology scales to multi-segment networks with ramp interactions, making it suitable for real-world deployment planning. The framework lays a solid foundation for learning-based traffic control solutions that are robust, interpretable, and efficient.

Future directions include:

- Integration with connected vehicle infrastructure
- Multi-agent coordination for network-wide optimization
- Real-world pilot testing with Dubai's RTA and other parts of the world

REFERENCES

- [1] B. Khondaker and L. Kattan, "Variable speed limit: an overview," *Transportation Letters*, vol. 7, no. 5, pp. 264-278, Jan. 2015.
- [2] E. Walraven, M. T. J. Spaan, and B. Bakker, "Traffic flow optimization: A reinforcement learning approach," *Engineering Applications of Artificial Intelligence*, vol. 52, pp. 203-212, May 2016.
- [3] M. Naem, S. T. H. Rizvi, and A. Coronato, "A Gentle Introduction to Reinforcement Learning and its Application in Different Fields," *IEEE Access*, vol. 8, pp. 209320-209344, 2020.
- [4] T. Haarnoja, A. Zhou, P. Abbeel, and S. Levine, "Soft Actor-Critic: Off-Policy Maximum Entropy Deep Reinforcement Learning with a Stochastic Actor," 2018.
- [5] I. Apamichail, "METANET Calibration," *IEEE ITS*, 2019.
- [6] Q. Yang, T. D. Simão, S. H. Tindemans, and J. Spaan, "WCSAC: Worst-Case Soft Actor Critic for Safety-Constrained Reinforcement Learning," *ResearchGate*, May 18, 2021.
- [7] P. Ševera, "The Ripple Effect of Traffic Congestion," *Goodvision-live.com*, Mar. 07, 2023.
- [8] Y. Zhang, I. I. Sirmatel, F. Alasiri, P. A. Ioannou, and N. Geroliminis, "Comparison of Feedback Linearization and Model Predictive Techniques for Variable Speed Limit Control," *21st International Conference on Intelligent Transportation Systems (ITSC)*, pp. 3000-3005, Nov. 2018.
- [9] J. Yang, P. Wang, F. Golpayegani, and S. Wang, "DVS-RG: Differential Variable Speed Limits Control using Deep Reinforcement Learning with Graph State Representation," *JOURNAL OF LATEX CLASS FILES*, vol. , no. 1, 2024.
- [10] M. Fares and huypham37, "Comparative Analysis of Reinforcement Learning Algorithms in Multi-Agent Soccer Environment," 2024.
- [11] Z. Zhou, Z. Yang, Y. Zhang, Y. Huang, H. Chen, and Z. Yu, "A comprehensive study of speed prediction in transportation system: From vehicle to traffic," *iScience*, 2022.
- [12] M. Robinson, "Examples of Variable Speed Limit Applications," *Transportation Research Board Annual Meeting*.
- [13] Haarnoja et al., "Soft Actor-Critic Algorithms and Applications," 2018.
- [14] J. Lu, "Reinforcement Learning for Freeway Variable Speed Limit Control: A Mixed Traffic Flow Case Study," Department of Civil and Environmental Engineering, University of California, Berkeley, Berkeley, CA 94720, USA, 2022.
- [15] Y. Zhang, M. Quinones-Grueiro, Z. Zhang, Y. Wang, W. Barbour, G. Biswas, and D. Work, "MARVEL: Multi-Agent Reinforcement-Learning for Large-Scale Variable Speed Limits," 2024.
- [16] B. Khondaker and L. Kattan, "Variable speed limit: an overview," *Transportation Letters*, vol. 7, no. 5, pp. 264-278, 2015.
- [17] M. Papageorgiou et al., "Real-time merging traffic control," *Transportation Research Part C*, vol. 18, pp. 807-820, 2010.
- [18] L. Li et al., "Reinforcement learning for variable speed limits control," *IEEE Trans. ITS*, vol. 21, no. 7, 2020.
- [19] "Dubai-Sharjah commute study," *Khaleej Times*, 2025.
- [20] A. Hegyi et al., "Model predictive control for optimal coordination of ramp metering and variable speed limits," *Transportation Research Part C*, vol. 13, pp. 185-209, 2005.